GITAM Deemed to be University

Civil Engineering | Course Code: CE304

Transportation & Highway Engineering - Laboratory Manual & Practical Gui

Syllabus Focus: CE304 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Highway Planning & Alignments

" Syllabus Scope: Highway development plans, Road patterns, Engineering surveys for highway location, Env

Unit 2: Geometric Design of Highways

" Syllabus Scope: Design speed, Cross-sectional elements, Camber, Sight distances SSD, OSD, Horizontal al

CE304: Transportation & Highway Engineering - Advanced Topics & Assessment

Academic Track: Civil Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Pavement Materials & Testing

" Subgrade soil characterization, CBR test, Crushing and Impact value of aggregates, Bitumen penetration, Duct

Unit 4: Pavement Design Principles

" Design of flexible pavements as per IRC:37 guidelines, Westergaard stress equations for rigid pavements, Tem

Unit 5: Traffic Engineering & Highway Maintenance

" Traffic volume, speed, and density studies, Level of Service LOS, Traffic signals design Webster method, Pav

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Aggregate Impact Value and Crushing Strength Testing for Highway Pavements

Lab 2: Penetration, Ductility, and Softening Point Characterization of Bituminous Binders

Lab 3: California Bearing Ratio CBR Laboratory Determination for Subgrade Soil

Lab 4: Traffic Volume and Speed Study at a Busy Urban Intersection with Signal Timing Design

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* S.K. Khanna, C.E.G. Justo - Highway Engineering Nem Chand & Bros 10th Ed

* Paul H. Wright, Karen Dixon - Highway Engineering Wiley 7th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students